Metrics: The Mean Absolute Error (MAE) is used to evaluate the performance of the boundary detection model. It measures the average absolute difference between the predicted position index and the actual changing point.

4 Task Organization

The shared task was run in two phases:

Development Phase. Only training and development data were provided to the participants, with no gold labels available for the development set. Participants competed against each other to achieve the best performance on the development set. A live leaderboard on CodaLab was made available to track all submissions. Teams could make an unlimited number of submissions, and the best score for each team, regardless of the submission time, was displayed in real time on CodaLab.

Test Phase. The test set was released, containing two additional languages—German and Italian for Subtask A Multilingual Track, generator GPT-4 for the Monolingual Track, and a new domain (student essays) for Subtask B. For Subtask C, both new domains and generators were introduced (GPT-4 and LLaMA-2 series based on PeerRead and OUTFOX), which were not disclosed to the participants beforehand (referred to as surprise languages, domains, and generators).

Participants were given approximately three weeks to prepare their predictions. They could submit multiple runs, but they wouldn't receive feedback on their performance. Only the latest submission from each team was considered official and used for the final team ranking.

In total, 125 teams submitted results for Subtask A Monolingual, 62 for Subtask A Multilingual, 70 for Subtask B, and 30 for Subtask C. Additionally, 54 teams submitted system description papers.

After the competition concluded, we released the gold labels for both the development and test sets. Furthermore, we kept the submission system open for the test dataset for post-shared task evaluations and to monitor the state of the art across the different subtasks.

5 Participating Systems

In this section, we first summarize common features for all teams based on the information they provided in the Google Docs. Then, we delve into the methods employed by the top 3 teams, accompanied by brief descriptions of the approaches utilized by the other top 10 teams.

Team Name	Ranking	small PLM	LLM	GPT	fine-tuning	zero-shot	few-shot (k=?)	Data augmentation	External Data
Genaios	1		✓						
USTC-BUPT	2	✓			✓				
petkaz	12	✓			✓				
HU	17		✓		✓			✓	
TrustAI	20	✓			✓				
L3i++	25		✓		✓				
art-nat-HHU	26	✓			✓				
Unibuc - NLP	28	✓			✓			✓	
NewbieML	30	✓							
QUST	31		✓		✓			✓	
NootNoot	39	✓			✓				
Mast Kalandar	40		✓		✓				
I2C-Huelva	41		✓		✓				
Werkzeug	45	✓			✓				
NCL-UoR	50		✓		✓				
Sharif-MGTD	51		✓		✓	✓			
Collectivized Semantics	62	✓			✓				
SINAI	61		✓		✓				
MasonTigers	71	✓	✓		✓	✓			
DUTh	73		✓		✓				
surbhi	74		✓		✓				
KInIT	77		✓		✓	✓			
RUG-D	100		✓		✓				✓
RUG-5	101	✓	✓		✓				
RUG-3	114		✓	✓	✓				
Mashee	115		✓				2		
RUG-1	117	✓							

Table 5: **Subtask A monolingual** participants methods overview. *small PLM*: Pre-trained Language Model is used, *LLM*: LLM is used, *GPT* indicates if any GPT models are used, *fine-tuning*: applying fine-tuned models, *zero-shot* and *few-shot (k=?)* that zero or more examples are used as demonstrations in in-context learning based on LLMs, *Data augmentation* and *External Data* referes to that augmented data or other external data have been used.

The approaches of all teams are presented in Appendix A.

5.1 Monolingual Human vs Machine

Table 5 provides a high-level overview of the methodologies employed by the top-ranking systems in Subtask A monolingual. Most systems utilized either a Pretrained Language Model (PLM) or a Large Language Model (LLM), with the majority of participants fine-tuning their models. Usage of GPT, external data, and few-shot methods was observed in only one team each.

Team Genaios$_{\text{STA_mono:1}}$ (Sarvazyan et al., 2024) achieved the highest performance in this subtask by extracting token-level probabilistic features (log probability and entropy) using four LLaMA-2 models: LLaMA-2-7B, LLaMA-2-7B-chat, LLaMA-2-13B, and LLaMA-2-13B-chat. These features were then fed into a Transformer Encoder trained in a supervised manner.

Team USTC-BUPT$_{\textbf{STA_mono:2}}$ (Guo et al., 2024) secured the second position. Their model is built upon RoBERTa, with the addition of two classification heads: one for binary classification (human or machine) using MLP layers, and another for domain classification (e.g., news, essays, etc.). The latter is equipped with an MLP layer and a gradient reversal layer to enhance transferability between the training and test sets. A sum-up loss is applied, resulting in approximately 8% improvement compared to the RoBERTa baseline.

Team PetKaz$_{\textbf{STA_mono:12}}$ (Petukhova et al., 2024) utilized a fine-tuned RoBERTa augmented with diverse linguistic features.

In addition to the top three teams: **Team HU$_{\textbf{STA_mono:17}}$** (Roy Dipta and Shahriar, 2024) employed a contrastive learning-based approach, fine-tuning MPNet on an augmented dataset. **Team TrustAI$_{\textbf{STA_mono:20}}$** ensembles several classical ML classifiers, Naive Bayes, LightGBM and SGD. **Team L3i++$_{\textbf{STA\_mono:24}}$** (Tran et al., 2024) investigated various approaches including likelihood, fine-tuning small PLMs, and LLMs, with the latter, fine-tuned LLaMA-2-7B, proving to be the most effective. **Team art-nat-HHU$_{\textbf{STA_mono:25}}$** (Ciccarelli et al., 2024) utilized a RoBERTa-base model combined with syntactic, lexical, probabilistic, and stylistic features. **Team Unibuc - NLP$_{\textbf{STA_mono:28}}$** (Marchitan et al., 2024) jointly trained Subtasks A and B based on RoBERTa. Most other teams fine-tuned either RoBERTa or XLM-RoBERTa for MGT detection, enhancing the models through various techniques, ranging from a mixture of experts by **Team Werkzeug$_{\textbf{STA_mono:45}}$** (Wu et al., 2024) to low-rank adaptation by **Team NCL-UoR$_{\textbf{STA_mono:50}}$** (Xiong et al., 2024), while **Team Sharif-MGTD$_{\textbf{STA_mono:51}}$** (Ebrahimi et al., 2024) preferred careful fine-tuning of PLMs alone.

5.2 Multilingual Human vs Machine

Table 6 provides an overview of the methods employed by the top-performing systems for Subtask A Multilingual. Various techniques are utilized, including zero-shot learning based on LLMs, PLM-based classifiers, and ensemble models.

Team Name	Ranking	small PLM	LLM	GPT	Fine-tuning	Zero-shot	Data augmentation	External Data
USTC-BUPT	1		✓		✓			
FI Group	2	✓			✓			
KInIT	3		✓		✓	✓		
L3i++	5		✓		✓			
QUST	6		✓		✓		✓	
AIpom	9		✓		✓			
SINAI	21		✓		✓			
Unibuc-NLP	22	✓			✓			
Werkzeug	30	✓			✓			
RUG-5	32	✓	✓		✓			
DUTh	33		✓		✓			
RUG-D	39		✓		✓			✓
MasonTigers	49	✓	✓		✓	✓		
TrustAI	55	✓			✓			

Table 6: **Subtask A multilingual** participants methods.

Team USTC-BUPT$_{\textbf{STA_Multi:1}}$ (Guo et al., 2024) secured the top position. They initially detect the language of the input text. For English text, they average embeddings from LLaMA-2-70B, followed by classification through a two-stage CNN. For texts in other languages, the classification problem is transformed into fine-tuning a next-token prediction task using the mT5 model, incorporating special tokens for classification. Their approach integrates both monolingual and multilingual strategies, exploiting large language models for direct embedding extraction and model fine-tuning. This enables the system to adeptly handle text classification across a diverse range of languages, especially those with fewer resources.

Team FI Group$_{\textbf{STA_Multi:2}}$ (Ben-Fares et al., 2024) implemented a hierarchical fusion strategy that adaptively combines representations from different layers of XLM-RoBERTa-large, moving beyond the conventional "[CLS]" token classification to sequence labeling for enhanced detection of stylistic nuances.

Team KInIT$_{\textbf{STA_Multi:3}}$ (Spiegel and Macko, 2024) combined fine-tuned LLMs with zero-shot statistical methods, employing a two-step majority voting system for predictions. Their method emphasizes language identification, per-language threshold calibration, and the integration of both fine-tuned and statistical detection methods, demonstrating the power of ensemble strategies. For the LLMs, they utilized QLoRA PEFT to fine-tune

Team Name	Ranking	small PLM	LLM	GPT	fine-tuning	zero-shot	Data augmentation
AISPACE	1		✓		✓		✓
Unibuc - NLP	2	✓			✓		✓
USTC-BUPT	3		✓				
L3i++	6		✓		✓		
MLab	7				✓		
Werkzeug	8	✓			✓		
TrustAI	14	✓			✓		
MGTD4ADL	17				✓		✓
scalar	18	✓					✓
UMUT	23	✓			✓		
QUST	36		✓		✓		✓
MasonTigers	38	✓	✓		✓	✓	
RUG-5	41	✓	✓		✓		
RUG-D	44		✓		✓		
DUTh	49		✓		✓		
clulab-UofA	62		✓	✓	✓		✓

Table 7: **Subtask B** Participants method overview.

Falcon-7B and Mistral-7B.

Other teams explored various approaches, like using LoRA-finetuned LLMs as classifiers (**Team AIpom**$_{\textbf{STA\_Multi:9}}$) (Shirnin et al., 2024), using semantic and syntactic aspects of the texts (**RFBES**$_{\textbf{STA\_Multi:10}}$) (Heydari Rad et al., 2024) or fusing perplexity with text and adding a classification head (**Team SINAI**$_{\textbf{STA\_Multi:21}}$) (Gutiérrez Megías et al., 2024). Each team's method provides insights into the complexities of multilingual text detection, ranging from the use of specific LLMs and PLMs to the use of linguistic and probabilistic metrics and ensemble techniques (Wu et al., 2024; Brekhof et al., 2024; Kyriakou et al., 2024; Puspo et al., 2024; Urlana et al., 2024).

5.3 Multi-way Detection

Table 7 provides an overview of the approaches employed by the top-ranking systems for Subtask B. Similar to Subtask A, most solutions do not use GPT and zero-shot approaches. The best-performing solutions primarily exploit LLMs and data augmentation.

Team AISPACE$_{\textbf{STB:1}}$ (Gu and Meng, 2024) achieved the highest performance in this subtask by fine-tuning various encoder and encoder-decoder models, including RoBERTa, DeBERTa, XLNet, Longformer, and T5. They augmented the data with instances from Subtask A and explored the effects of different loss functions and learning rate values. Their method leverages a weighted Cross-Entropy loss to balance samples in different classes and uses an ensemble of fine-tuned models to improve robustness.

Team Unibuc - NLP$_{\textbf{STB:2}}$ (Marchitan et al., 2024) employed a Transformer-based model with a unique two-layer feed-forward network as a classification head. They also augmented the data with instances from the Subtask A monolingual dataset.

Team USTC-BUPT$_{\textbf{STB:3}}$ (Guo et al., 2024) leveraged LLaMA-2-70B to obtain token embeddings and applied a three-stage classification. They first distinguished human-generated from machine-generated text using LLaMA-2-70B, then categorized ChatGPT and Cohere as one class and distinguished them from davinci-003, BLOOMz, and Dolly-v2. Finally, they performed binary classification between ChatGPT and Cohere.

Team L3i++$_{\textbf{STB:6}}$ (Tran et al., 2024) conducted a comparative study among three groups of methods: metric-based models, fine-tuned classification language models (RoBERTa, XLM-R), and a fine-tuned LLM, LLaMA-2-7B, finding LLaMA-2 to outperform other methods. They analyzed errors and various factors in their paper.

Team MLab$_{\textbf{STB:7}}$ (Li et al., 2024) fine-tuned DeBERTa and analyzed the embeddings from the last layer to provide insights into the embedding space of the model.

Team Werkzeug$_{\textbf{STB:8}}$ (Wu et al., 2024) utilized RoBERTa-large and XLM-RoBERTa-large to encode text, addressing the problem of anisotropy in text embeddings produced by pre-trained language models (PLMs) by introducing a learnable parametric whitening (PW) transformation. They also used multiple PW transformation layers as experts under the mixture-of-experts (MoE) architecture to capture features of LLM-generated text from different perspectives.

Other teams explored various approaches, including different loss functions and sentence transformers (**Team MGTD4ADL**$_{\textbf{STB:17}}$) (Chen et al., 2024), RoBERTa fine-tuning (**Team UMUTeam**$_{\textbf{STB:23}}$) (pan et al., 2024), stacking ensemble techniques (**Team MasonTigers**$_{\textbf{STB:38}}$) (Puspo et al., 2024), and basic ML models with linguistic-stylistic features (**Team RUG-5**$_{\textbf{STB:41}}$)